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Review

Review on microaeration-based anaerobic digestion: State of the art, challenges, and prospectives

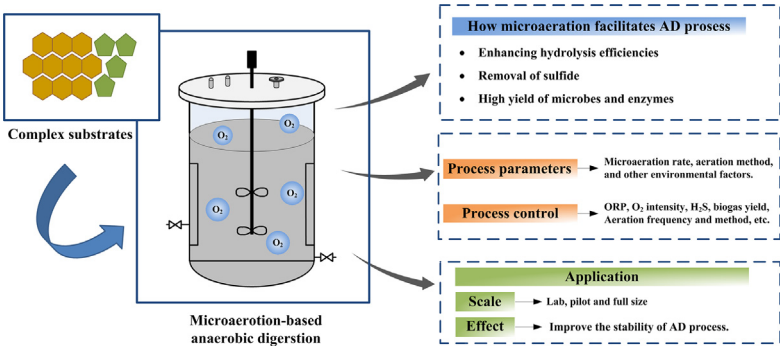
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HIGHLIGHTS

- Microaeration in AD system facilitates processing efficiency and system stability.
- The presence of oxygen leads to a high yield of facultative acidogens and excreted enzymes.
- Microaeration rate is the foremost parameter connected to the AD performance.
- The combination of aeration and AD has been applied in full-scale implementation successfully.

GRAPHICAL ABSTRACT



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ABSTRACT

Microaeration (dosing small quantities of air or oxygen) is an effective approach to facilitate anaerobic digestion (AD) process and has gained increased attention in recent years. The underlying mechanisms of the facilitation effect of microaeration on AD process were reviewed in terms of accelerating hydrolysis, scavenging hydrogen sulfide, and affecting microbial diversity. Process parameters and control strategies were summarized to reveal considerable factors in implementing microaeration-based AD process. In addition, current applications, including lab-, pilot- and full-scale level cases, were summarized to provide guidance for further improvement in large-scale applications. The challenges and future perspectives were also highlighted to promote the development of AD process associated with microaeration.

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1. Introduction

With the increase in concerns about environmental protection and sustainable development, the treatment of waste has gained considerable attention in recent years. The focus has shifted from disposal effect into cost-effective and environment-friendly techniques (Robles et al., 2018). Among all existing strategies, anaerobic digestion (AD), which is a fermentation process by microorganisms in the absence of oxygen (O_2), is a mature technique for the decomposition of organic waste materials and wastewater (Karthikeyan and Visvanathan, 2013; Wu et al., 2019). Compared with conventional aerobic processes, AD has the superiorities of less energy consumption, biogas recovery, and lower sludge production (Bae et al., 2014; McCarty et al., 2011; Stazi and Tomei, 2018). AD process has gained considerable attention and has been used extensively to treat organic waste due to the production of renewable energy source in the form of biogas (containing mainly methane and carbon dioxide), which is considered a substitute for fossil fuels (Curry and Pillay, 2012; Pellera and Gidarakos, 2017; Rajagopal et al., 2013; Sun et al., 2016).

Despite the above-mentioned merits, AD process has various defects, such as low hydrolysis rate, release of toxic hydrogen sulfide (H_2S), and poor system stability, because of its sensitivity to the changes in operation and substrate conditions. The microbial consortia can suffer from high concentration of volatile fatty acids (VFAs) produced in the anaerobic biochemical processes; this condition is closely related to the growth and activity (Lim et al., 2014; Nguyen et al., 2015). Microaeration treatment technology, which has attractive advantages of accelerating hydrolysis, scavenging hydrogen sulfide, preventing massive accumulation of VFAs, and high biogas yield, has been suggested as an effective measure to improve the stability of the anaerobic system for helping to solve the aforementioned problems (Khanal and Huang, 2006; Nguyen and Khanal, 2018; Ramos and Fdz-Polanco, 2013).

Theoretically, oxygen or air to the AD process is forbidden because of the inhibition in growth and metabolism of obligate anaerobic microorganisms, especially methanogens. The gas mixture of methane and O_2 also has a risk of explosion (Nguyen and Khanal, 2018; Ruan et al., 2019). However, recent studies have proven that dosing small quantities of oxygen or air can be used as a strategy to improve the stability of AD process. Instead of causing any inhibition, the microaeration-based anaerobic process enhances anaerobic performance (Fu et al., 2016; Girotto et al., 2018). Several comprehensive reviews are currently assessing the impacts of microaeration on the anaerobic treatment (Botheju and Bakke, 2011; Girotto et al., 2018; Jenicek et al., 2010; Nguyen and Khanal, 2018). According to previous reports, a specific environment that overlaps between anaerobic and aerobic conditions is created by the microaeration procedure, which maintains niches for anaerobes and facultative anaerobes in such surrounding. The presence of oxygen leads to a high yield and population of facultative acidogens and results in a high quantity of excreted enzymes.

The microaeration-based AD process has been widely applied to treat organic waste and improves processing efficiency and system stability (Lim et al., 2014). The essential mechanism of enhancing AD process via exposing oxygen is the augmentation of activity of facultative hydrolytic and acidogenic microorganisms under microaerobic conditions (Lim and Wang, 2013; Peces et al., 2016; Xu et al., 2014).

This study critically summarized the important aspects of microaeration-based AD. The beneficial effects of microaeration on hydrolysis process, sulfide removal, and microbial community structures were critically reviewed. Important considerations for process control strategies were reviewed as well. Current applications were summarized to provide guidance for further applications. Challenges and future perspectives on microaeration-based anaerobic system were also highlighted.

2. Microaeration-based AD

2.1. Definition of microaeration

Microaeration is a system with low concentration of dissolved oxygen (DO) between aerobic and anaerobic environments, and this system is constructed by dosing small quantities of air or oxygen (DO range of 0.1–1.0 mg/L) into the anaerobic bioreactor (Hasegawa et al., 2000; Wu et al., 2015; Chen et al., 2017; Nguyen and Khanal, 2018). Microaeration is referred to in different terminologies, such as limited aeration, micro-oxygenation, oxygenation, or moderate oxygenation. The terms “microaerobic” and “microaerophilic” are also used to represent the reactor condition (Krayzelova et al., 2015).

Unlike aerobic system in which DO can be measured online by electrode, the O_2 concentration in microaerobic process is too low to measure, and pH detection is insensitive to slight changes in intracellular metabolism. A valid alternative is to measure the oxidation–reduction potential (ORP), which reflects the overall electron transfer and net balance of reducing equivalents in intracellular metabolism. ORP value varies linearly with the logarithmic of O_2 concentration. Therefore, the ORP electrode can precisely sense very small changes in O_2 concentration in aqueous phase.

Microaeration has been defined with the ORP value that ranges from 0 mV to −300 mV as determined by standard hydrogen electrode. The ORP level of anaerobic environment is −300 mV or lower, while that

of aerobic condition is higher than 0 mV (Nguyen and Khanal, 2018). However, several ORP values not in this range have also been reported during microaeration, such as < −460 mV (Duangmanee et al., 2007) and from −320 mV to −270 mV (Nghiem et al., 2014). This variation is probably determined by the particularity of each reactor and operating conditions (Krayzelova et al., 2015).

2.2. Microaeration formation

The microaeration environment can be realized by several aerobic treatment measures. Microaeration intensity (or rate), which measured in terms of aeration or oxygen flow rate, is the main factor determining the effects of microaeration on AD process (Nguyen and Khanal, 2018). Various aeration modalities have been performed by air or oxygen injection, electrolytic aeration, and even natural aeration (leaving the tested configuration open with or without shaking). The injection schemes in anaerobic digesters mainly include dosage in liquid phase, gas phase, and biogas recirculation avenue (Krayzelova et al., 2015). The aeration frequencies can be one time, intermittently, or continuously depending on practical requirements (González-González and Cuadros, 2015; Girotto et al., 2018; Rafieenia et al., 2016).

Air or oxygen can be dosed at different stages of AD, including pretreatment, during the process, and in the latter stages of AD process depending on different purposes. Table 1 provides a comparison of microaeration operations in different AD phases. When microaeration is performed as a pre-treatment method, the aeration intensity should be given careful consideration because the excess oxygen amount reduces the methane potential, and insufficient oxygen amount potentially fails to produce any effect (Girotto et al., 2018).

The approaches implemented to aerobically stimulate AD performance include microaeration or limited oxygen injection throughout the entire AD process. Microaeration is used throughout the entire AD process to improve stability and performance of the AD system and enhance the methane yield.

2.3. Metabolism in microaeration-based AD

The unique microaerobic environment provides niches for both anaerobes and micro-aerobes. The modeling of biological and chemical processes is regarded as importance for metabolic pathway analysis, control and optimization (Donoso-Bravo et al., 2018).

The most widely applied AD model is the IWA ADM1; the organic biodegradation in this system is divided into five procedures, namely, disintegration, hydrolysis, acidogenesis (fermentation), acetogenesis (acetate generation), and methanogenesis (methane generation) (Batstone et al., 2002; Botheju and Bakke, 2011). Among all the steps, hydrolysis is the rate-limiting step because it involves enzyme-mediated chemical–biochemical conversion of long-chain polymeric organic compounds (e.g., polysaccharides, protein, and fats) into simple and soluble organics (e.g., monosaccharides, amino acids, and long-chain fatty acids) that can be utilized by microbes during the subsequent acidogenesis stage (Song et al., 2019).

Under aerobic condition, acetyl coenzyme A goes through tricarboxylic acid cycle to be completely oxidized to CO_2. Fig. 1 shows the biological metabolic process of organic carbon in anaerobic and aerobic environments; this process is adapted from Madigan et al. (2015), Nguyen and Khanal (2018), and Robles et al. (2018).

Recent studies have developed several models for the microaerobic process in AD. Botheju et al. (2009) developed a model describing oxygen effects in anaerobic digestion, highlighting the improvement of organic matters' solubility and aerobic oxidation of soluble carbon. The inhibition of obligatory anaerobic organisms was also concerned in the model. Krayzelova et al. (2017) extended ADM1 with sulfate reduction and sulfide oxidation to optimize the microaeration treatment of sulfate-rich wastewater. The generated model ADM1-S/O, which validated against experimental data, shows a good description of H_2S concentrations in the biogas.

3. Facilitation effect of microaeration on AD process

Microaeration improves the overall stability of AD process (Nguyen and Khanal, 2018; Khanal and Huang, 2006; Ramos and Fdz-Polanco, 2013). The underlying mechanisms of the facilitation effect of microaeration on AD process are summarized as follows.

3.1. Enhancing hydrolysis efficiencies

As reported in previous studies, enhancing hydrolysis accelerates the dissolution of complex substrates into simple organic matters for the subsequent degradation (Cesaro and Belgiorno, 2014; Guo et al., 2015; Mathews et al., 2015; Shrestha et al., 2017). Microaeration effectively promotes hydrolysis of different substrates, such as municipal wastewater (Zhang et al., 2019b), municipal solid waste (Nguyen et al., 2007), food wastes (Xu et al., 2014), vegetable and flower wastes (Zhu et al., 2009), olive mill wastewater (González-González and Cuadros, 2015), and sewage sludge (Montalvo et al., 2016a, 2016b). Exoenzyme activities of cellulase and protease are observed after 5-day pre-aeration, in which hydrolysis is apparent. In the presence of O_2, a high production of facultative hydrolytic and acidogenic microorganisms, which impact the hydrolysis of organic waste, are obtained and high amounts of enzymes are excreted. Hydrolysis reaction is driven by extracellular enzymes (e.g., amylase, protease, and lipase) excreted by acidogens; thus, a high yield of acidogenic biomass leads to increased hydrolysis (Botheju and Bakke, 2011).

Microaeration not only accelerates hydrolysis and acidogenesis but also impacts the product patterns (Lim et al., 2014). As the intermediate metabolites form during the acidogenic step in AD system, VFAs are precursors to form biogas via methanogenesis (Agler et al., 2011). VFAs can also serve as organic carbon sources to accelerate the removal of biological nutrients (Lee et al., 2014). Thus, VFAs are produced and consumed simultaneously in one system. In the presence of O_2, VFAs, such as propionic, butyric, valeric, and lactic acids, are transformed into acetic acid efficiently (Lim and Wang, 2013). Microaeration, which is an efficient treatment method, promotes the growth, activity, and diversity of rapid-growing facultative microbes and excretion of enzymes to enhance hydrolysis of AD and maintain a low VFA accumulation concentration.

Table 1
Comparison of microaeration operations in different AD phases.

Operational schemes	Aeration Method	Consequence	Possible drawbacks	Reference
Pre-aeration	Natural aeration; Air/O_2 injection Electrolytic aeration	Improve the first AD step, hydrolysis	Loss of VS and lower methane production	Mshandete et al., 2005; Fu et al., 2015; Botheju and Bakke, 2011
Microaeration during AD process	Air/O_2 injection into the reactor	Reduce sulphides content and minimize the toxic effect of aqueous sulphides	Negative effect on methane generation	Botheju et al., 2010; Girotto et al., 2018
In the late AD Stage	Air/O_2 injection	Improve both methane yield and digestate biostability	Reduce *E.coli* population	Fu et al., 2015; Tomei et al., 2016

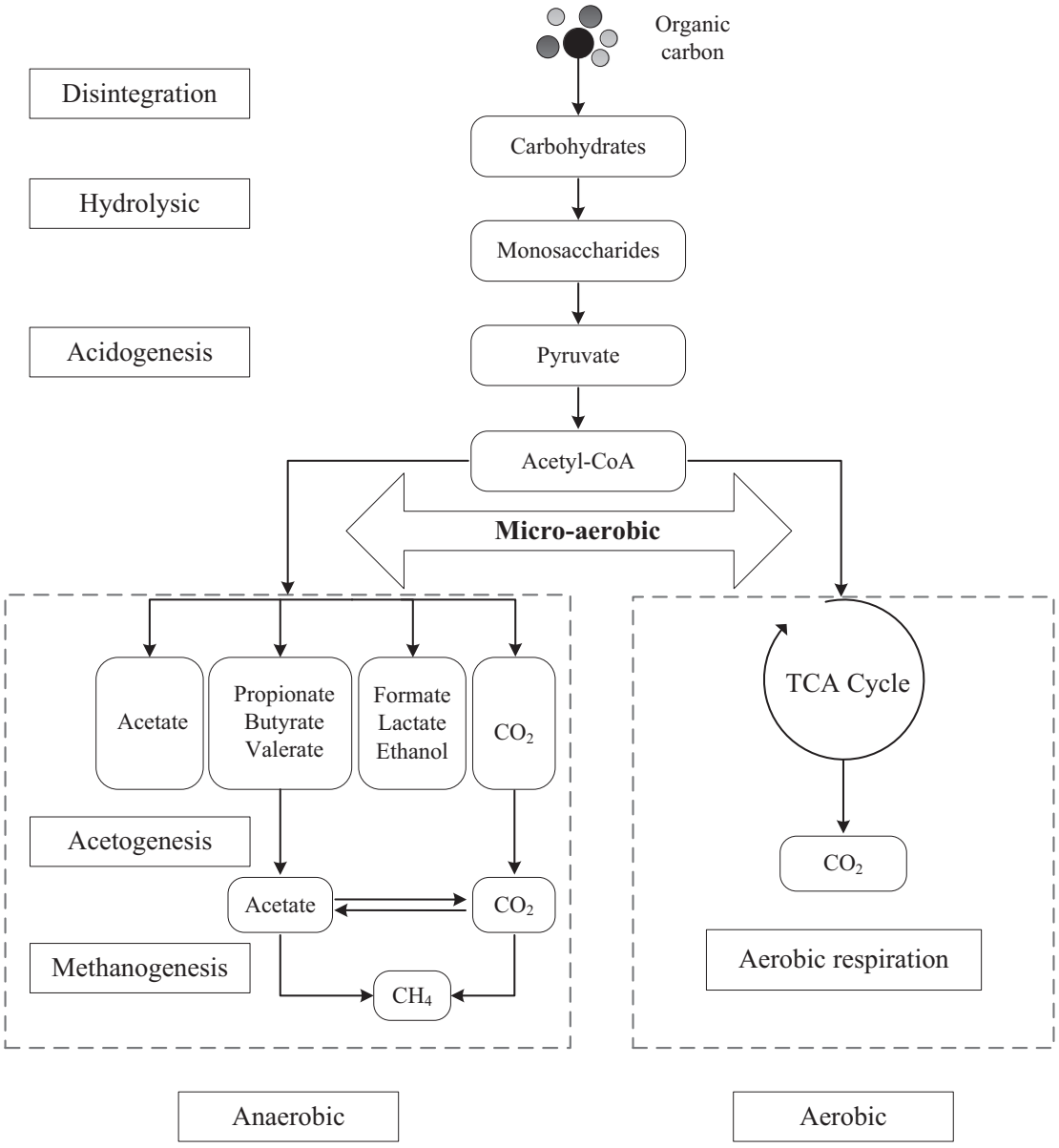


Fig. 1. Biological metabolic process of organic carbon in microaerobic environment. TCA: tricarboxylic acid cycle.

3.2. Removal of sulfide

During AD process, sulfur compounds in the substrate are reduced to hydrogen sulfide in biogas and sulfide (HS^- and S_2^-) in liquid phase. H_2S level can reach as high as 10,000 ppm (Wellinger and Linberg, 2000). High concentration of odorous H_2S in biogas reduces its reuse potential and economic value and causes corrosion of concrete and steel pipeline. Moreover, sulfide in liquid phase is not only toxic to methanogens but also destructive to the AD system (Nghiem et al., 2014; Krayzelova et al., 2015).

Direct sulfide removal by dosing of O_2 or air to anaerobic bioreactor is a simple and effective technique in cases of treating sulfate-rich substrates (Krayzelova et al., 2015; Ramos et al., 2014b; Zitomer and Shrout, 2000) and low-sulfate wastewater (van der Zee et al., 2007). Compared with other physical, chemical, and biological techniques for desulfurization of biogas, microaeration has high removal efficiency (up to 99%) and low operating costs and slightly modifies the AD process (Chen et al., 2010; Díaz et al., 2011a, 2011b; Ramos et al., 2014c).

The oxidation of sulfide, which involves biological and chemical processes, must be analyzed to clarify the mechanism of oxygen dosage on sulfide removal. The major reactions and energy generation contained in sulfide removal under aerobic conditions are given as follows (Buisman et al., 1990; Janssen et al., 1995; Kuenen, 1975):

$$2HS^- + O_2 \rightarrow 2S^0 + 2OH^- \quad \Delta G^o = -169.35\,\mathrm{KJ/mol} \tag{1}$$

$$2HS^- + 4O_2 \rightarrow 2SO_4^{2-} + 2H^+ \quad \Delta G^o = -732.35\,\mathrm{KJ/mol} \tag{2}$$

$$2HS^- + 2O_2 \rightarrow S_2O_3^{2-} + H_2O \quad \Delta G^o = -387.35\,\mathrm{KJ/mol} \tag{3}$$

The formation of polysulfides S_n^{2-} is the first step of sulfide oxidation and can be further protonated to form elemental sulfur (Krayzelova et al., 2015). During the biological process of desulfurization, oxygen is the terminal electron acceptor, while sulfide acts as the electron

donor. The production types of sulfur or sulfate depend on the amount of oxygen dosing. Under microaerobic conditions with low oxygen concentrations (<0.1 mg/L), sulfur is the primary oxidation product (Eq. (1)) and causes a segmental oxidation to thiosulfate (van den Ende and van Gemerden, 1993). In case of sulfide limiting, sulfate is produced in quantity of oxygen (Eq. (2)). Eq. (3) shows the formation of thiosulfate via oxidation of sulfide (Janssen et al., 1995). The sulfide oxidation in autotrophic denitrification bioreactor under microaeration environment may occur in three pathways. One is anaerobic pathway of sulfide oxidizing coupled with nitrate reduction. The others are aerobic pathways, in which oxygen acts as electron acceptor and oxidizes sulfide to sulfur, thiosulfate, or even to sulfate (Chen et al., 2013; Zhang et al., 2019a). The end-products of sulfur and sulfate can be regulated by the supplied quantity of oxygen (Munz et al., 2009).

Although dissolved sulfide in liquid phase can be obliterated over 99% in microaeration-based AD process, the issues caused by sulfur element precipitation and yield reduction of methane owing to aerobic oxidation still remain. The above-mentioned cases caused by surplus aeration will be the focus of biological desulfurization via microaeration in further studies.

3.3. Effect on microbial diversity

AD is the totality of the collective interaction of at least three main microbial groups, namely, acidogens (acid generating biomass), acetogens (acetate generating biomass), and methanogens (methane generating biomass), as classified according to the function of microorganisms. The microbial consortia under microaeration environment contain strict and facultative anaerobes as classified by the relationship between microbial growth and oxygen. With oxygen limitation, the facultative microorganisms are suggested to retain strict anaerobes by maintaining low redox potential and supplying increased growth factors (Gerritse et al., 1990; Kato et al., 2004; Li et al., 2014; Prit and Lee, 1983).

Microbes in AD system have considerable oxygen tolerance because of several mechanisms. Under microaerobic conditions, facultative microbes consume and partially reduce O_2 molecules for generating reactive oxygen species (i.e., O^{2-}, ·OH, and H_2O_2,). Accordingly, obligate anaerobes in anaerobic environment may survive and maintain activity under limited aeration with no or slight negative effects (Brioukhanov and Netrusov, 2007; Sheets et al., 2015). The essential mechanism is that rapid oxygen consumption capacity of facultative fermentative organisms can protect strict anaerobes by scavenging DO (Hao et al., 2009; Kato et al., 1993).

Fu et al. (2016) investigated the impact of microaeration on the microbial community structure. The relative abundances of phylum *Firmicutes*, class *Clostridia*, and order *Clostridiales*, which are related to hydrolysis, are increased in microaerobic environment compared with that under anaerobic condition. Moreover, the relative abundances of oxytolerant *Methanosarcina* and *Methanobacterium* are doubled in microaerobic environment. Meanwhile, specific methanogenic activity improves slightly. The evolution of microbial communities may be the reason behind the improvement effect of microaeration on anaerobic treatment efficiency. The VFA yield increase of nearly 330% is probably because of the incremental growth and metabolic activity of facultative hydrolytic and acidogenic bacteria after 4 days of micro-oxygenation, as reported by Lim and Wang (2013). The unexpected dominant aerobes of *Acetobacter peroxydans* are discovered in the AD system; this condition is due to that the reactor may be unknowingly exposed to partial aeration.

In AD systems, methane is the most desirable economic product produced by methanogens. However, H_2S formed by sulfate-reducing bacteria is an inhibitor on methanogenesis. Microaeration prevents the imbalance may be due to the hydraulic overload by removing sulfide and accelerating growth of hydrogenotrophic methanogens, which generate feedback to promote acetic formation and facilitate to keep pH constant (Ramos and Fdz-Polanco, 2013). Buettner and Noll (2018) investigated and reported that bacteria of high abundance in sewage treatment plants are *Proteobacteria*, *Firmicutes*, *Chloroflexi*, and *Bacteroidetes*. Methanogenic archaea, which is known as the strictest anaerobes, has slight to no tolerance to oxygen (Lagerkvist et al., 2015).

Small dosing of O_2 accelerates the diversity and activity of hydrolytic and fermentative microorganisms in AD process. Improvement of growth and metabolism of these bacteria is the basis of regulating and controlling the VFA concentration (either promote or demote), which results in the promotion of overall stability of the AD process.

4. Process control

4.1. Process parameters

Various factors can lead to the instability of the AD process because it contains complicated chemical and biological reactions. Among these factors, oxygen dosing rate (or microaeration rate) is the foremost parameter connected to the performance of digestion. A variable oxygen rate is requisite in different cases depending on the targets, such as enhancing hydrolysis, increasing VFA production, preventing VFA accumulation, removing hydrogen sulfide, improving methane yield, and stabilizing the system (Nguyen and Khanal, 2018).

An appropriate microaeration rate can significantly enhance the hydrolysis of substrates and the yield of products (Díaz et al., 2013). Xu et al. (2014) reported that optimal microaeration rate (258 L air/kg TS/d) increases the hydrolysis efficiencies of carbohydrates and proteins by 21%–27% and 38%–64%, respectively. In addition, more than three folds of intermediate products (e.g., acetic acid and butyric acid) are produced owing to the quickened acidogenesis as compared to that produced in anaerobic treatment. Wu et al. (2015) showed that limited aeration with DO from 0.2 mg/L to 0.3 mg/L is effective in improving the biodegradation of petrochemical wastewater and reducing the generation of H_2S. The amount of VFAs improves under optimal DO condition but declines significantly with the increase in DO concentration. Lim et al. (2014) obtained similar consequences in their research on co-digestion brown water and food waste by microaeration. Hao et al. (2009) also discovered that different dosing rates of O_2 have varying effects on hydrolysis in the potential of the process.

Apart from microaeration rate, the parameters of oxygen transfer rate (OTR) and oxygen utilization rate (OUR) should be concerned and pondered as references to provide sufficient oxygen intensity. The OTR depends on several parameters, such as reactor configuration, dosing method, and total solids of substrates in the reactor (Garcia-Ochoa and Gomez, 2009). The OUR depends on substrates and inoculum types supplied for the system (Nguyen and Khanal, 2018). The effects of inoculum type, oxygen dosage, and incubation time on VFA production in digestion of Napier grass were investigated by Sawatdeenarunat et al. (2017). The experimental group inoculated anaerobically by digested cattle manure gains a 13-fold higher production of VFAs than the group inoculated anaerobically by activated sludge. Additional carbon sources increase the degradation rates of nitrogen (61% to 83%) and phosphorus (61% to 97%) during anaerobic–aerobic wastewater treatment (Zheng et al., 2009). Nguyen and Khanal (2018) provided guidance for determining the microaeration rate at particular reactor conditions and for different purposes.

Microaeration rate, aeration method, substrate type, inoculum types, and other environmental factors (e.g., temperature and pH) are important affecting factors of the efficiency of microaeration-based AD system. Parameter optimization is important in promoting the efficiency and stability of AD.

4.2. Process control

Process control is quite important for precise control and adjustment in microaeration-based AD system. As discussed in the previous section,

dosing rate of air or O_2 is the foremost affecting factor of the performance of AD process. Selection between air and pure oxygen for dosing into anaerobic reactors can be determined by the digestion purposes and relevant cost. Precise control of O_2 dosing should be comprehensively investigated to maximize the efficiency of digestion. The ORP is sensitive to the existence of O_2 and linearly varies depending on the O_2 concentration in an aqueous phase. Thus, the tactic of ORP-based control has been recommended (Chang et al., 2014; Nghiem et al., 2014).

The use of ORP as a control parameter can maintain sufficient residual oxygen without inhibiting obligate anaerobes because dosing of oxygen can be controlled precisely depending on the oxygen consumption rate of facultative bacteria in the system. Khanal and Huang (2003a, 2003b, 2006) & Khanal et al., 2003 developed an ORP control system to regulate O_2 quantity for realizing the online sulfide oxidation. During oxygenation at ORP that ranges from −230 mV to −180 mV, the dissolved and gaseous sulfides were completely eliminated and the methane yield was increased to 56.3%. These studies have proven the reliability for achieving consistent online sulfide control by ORP regulated oxygenation. Liu et al. (2013) reported the influence of redox potentials on gene expression, protein metabolism, and ORP control strategies because ORP can reflect biochemical reactions. Nghiem et al. (2014) evaluated the effect of ORP on regulating the dosing of a minute amount of O_2 into anaerobic digester for increasing H_2S removal efficiency in biogas.

In addition to using ORP as a controlled variable and O_2 flow rate as manipulated variable, some other parameters are considered to improve the performance. Ramos and Fdz-Polanco (2014) have successfully used the biogas production and H_2S concentration as variables to control the rate of O_2 injection so as to control the biogas sulphide content. The study revealed that biogas yield could be applied to develop control strategies under variable organic load and stable sulfur load, while H_2S content was employed instead under unstable sulfur load condition.

The control strategies based on variables of aeration frequency and method are influential on AD process. A variable oxygen rate is required in most cases because the feed composition or rate variations will result in varying products. On account of low solubility of O_2, the strategy of intermittent aeration has been applied in previous studies. Khongsumran et al. (2014) supplied oxygen to the anaerobic reactor for 5 min every 2 h and achieved the highest hydrolysis efficiency of cellulose at the rate of 3.0 mL O_2/LR/d. Jagadabhi et al. (2010) demonstrated that dosing at low flow rates (10 LO_2/kgVS/min) during the first 11 days of AD results in improvement in VFA production but no significant increase in COD solubilization. Similarly, a quadruplicated VFA production and an improvement in conversion of the produced are also obtained with air introduced into an LBR.

Process monitoring and control based on the main physiochemical parameters is difficult to implement and have several limitations. For instance, the parameters cannot predict the following performance, and some parameters cannot be monitored online. This condition delays the process diagnosis. Understanding the composition and behavior of microbial consortia is beneficial to improving performance to some extent (Li et al., 2018).

5. Application of microaeration-based AD system

The combination of aeration and AD has been applied in several works to improve hydrolysis and increase substrate conversion efficiencies. Microaeration also can prevent the inversion of H_2S and oxidize existing H_2S to elementary sulfur (Giordano et al., 2014; Xue et al., 2015; Zhou et al., 2014). The process performances of microaeration in recent researches are summaried in Table 2. Positive effects and potential drawbacks in selected lab-, pilot- and full-scale applications are discussed as follows.

5.1. Lab- and pilot-scale cases

The studies of lab- and pilot-scale microaeration-based AD processes on improving the efficiency of organic degradation, reducing H_2S content and increasing methane yield were reported in a large number. Several selected and representative researches are discussed in this part.

Nghiem et al. (2014) investigated the effect of microaeration on reducing H_2S concentration in biogas during AD. The results showed that hydrogen sulfide in biogas in the microaerobic system decreases from over 6000 ppm to 30 ppm, while hydrogen sulfide in the reference digester (without micro-oxygen injection) is over 4000 ppm. However, no evident changes are observed in the VS and COD removal rates, pH, and alkalinity between microaerobic and anaerobic conditions. This phenomenon is probably due to the competitive oxygen utilization by sulfur oxidizing bacteria. In addition to the above-mentioned conclusion, the study confirmed the feasibility of regulating oxygen dosing by using ORP implemented at a scale of 50 L.

Xu et al. (2014) investigated the effect of different microaeration rates on the solubilization of food waste. The study verified that hydrolysis performance in inadequate aeration is similar to that in anaerobic reactor. However, adequate microaeration intensity (258 and 387 L-air/kg TS/d) significantly enhances the carbohydrate and protein hydrolysis by 21%–27% and 38%–64%, respectively. Appropriate microaeration rate promotes the hydrolysis of solid organic waste and methane yield without undesirable carbon loss.

Fu et al. (2016) achieved a 16.5% improvement in methane yield during the thermophilic AD of corn straw by injecting air into the bottles of this material with a syringe (2.63 mLO_2/LR/d). Chen et al. (2017) confirmed that the DO significantly increases the removal rate of sulfide and effectively inhibits the sulfide rebound in batch tests.

Niu et al. (2016) investigated the effect of DO in a microaerobic hydrolysis sludge in situ reduction process. The results showed that the sludge reduction efficiency increases from 42.9% to 68.3% when DO concentration in microaerobic tank decreases from 2.5 mg/L to 0.5 mg/L. The results revealed that microaeration in the sludge process reduction activated sludge (SPRAS) improves hydrolysis efficiency and enriches fermentative and predatory bacteria responsible for sludge reduction.

Zhang et al. (2019b) designed a novel reactor that combines falling water aeration and external reflux upflow microaerobic sludge to treat wastewater with low COD/TN ratio. The DO concentration can be accurately controlled by adjusting the reflux ratio of oxygenated water. At the reflux ratio of 5:1, the removal rates of NH_4^+ and TN reach 90.53% and 80.77%, respectively.

5.2. Success at full-scale implementation

Although application in full-scale of AD system is more complicate to operate, several implementations of microaeration-based AD process in full-scale has been reported in recent studies. Jenicek et al. (2008, 2010, 2014, 2017) reported a series of full-scale demonstrations of reducing H_2S concentration in biogas by microaeration. The work described the experiences obtained from 7 microaerobic digesters in the operation periods from 2003 to 2015 in central Europe. The desulfurization efficiencies of >90% were obtained in most of the cases. Moreover, microaeration could improve the degradability of COD and volatile suspended solids.

Kobayashi et al. (2012) selected a full-scale anaerobic digester with volume of 338 m^3 to investigate the microbial community for S^0 accumulation in headspace process. The microaerobic environment made it possible for utilizing the oxygen to oxidize H_2S retaining in the headspace by sulfide-oxidizing bacteria. During the phase with air addition, H_2S concentration was 1120 ± 670 ppm on average, while that was from 2000 to 4000 ppm during the phase without air addition. The result showed that air addition provided 68.2% reduction of H_2S in comparison with the control phase.

Table 2
Process performance of microaeration-based AD process in published studies.

Scale	Reactor (Volume)	Substrate	Micro-aeration intensity	Positive effect	Reference
Lab scale	Continuous-flow reactor (2.5 L)	Organic sludge	DO of 0–0.1 mg/L	Approximately 40% of the organic sludge solubilized after its treatment for 1–2 days.	Hasegawa et al., 2000
	Batch reactor (0.5 L)	Primary sludge	Air (~500 mL/d) added intermittently 15 s every minute	Enhance hydrolysis by 50–60%.	Johansen and Bakke, 2006
	UAF + SOU (4.5 L + 2 L)	wastewater	ORP controlled (−275 to −265 mV)	Sulfides eliminated completely; improvement in methane yield by 46%.	Khanal and Huang, 2006
	UASB (11*L*)	Evaporator condensate	Aeration rate of 0–5.5 mL/L/min	The removal rate of COD increased from 40% to 80%	Zhou et al., 2007
	LBR (1 L)	Grass-silage	Micro-aeration at flow rate of 1 L/min	Production of VFA increased 4-fold.	Jagadabhi et al., 2010
	Erlenmeyer flask (1.6 L)	Primary sludge	Micro-aeration rate of 0.00156 vvm effectively COD	Increase solubilization of COD by hydrolysis of carbohydrates and protein.	Diak et al., 2013
	CSTR (0.25 L)	Food waste, yellow Water and brown water	Microaeration pretreatment of 37.5 mL O_2/L_R-day	Enhance methane yield by 21% when pretreatment was applied to inoculated substrates.	Lim and Wang, 2013
	UASB (2.7 L)	Synthetic brewery wastewater	Air amount of 1 L/d	Elemental sulfur produced by microaeration did not accumulate in granular sludge.	Krayzelova et al., 2014
	LBR + UASB (4.6 L + 10 L)	Food waste	129, 258, 387 L air/kg TS/d with a flow rate of 1 L/min	Enhance the hydrolysis of carbohydrate and protein by 38–64%; produce >3-fold of acetic acid and butyric acid.	Xu et al., 2014
	Digestion reactor (4.5 L)	Petrochemical wastewater	DO from 0.2 to 0.3 mg/L (average ORP of −210 mV)	Increase BOD_5/COD from 0.23 to 0.43; almost no H_2S detected in the off-gas.	Wu et al., 2015
	Batch reactor (0.2 L)	Corn straw	Air loads of 0, 12.5, 25, 50, and 100 mL/L_R-d	Maximum cumulative methane yield of 216.8 mL/g VSsubstrate and maximum VS removal efficiency of 54.3%.	Fu et al., 2016
	Glass reactor (5 L)	Sewage sludge	Air flow rate of 0.3 vvm	Enhance methane yield by 211%.	Montalvo et al., 2016a
	Batch reactor (5 L)	Mixed sludge	Air flow rate of 0.35 vvm	Methane production increased 114%.	Montalvo et al., 2016b
	EGSB (1)	Municipal Wastewater	DO of 0.2 mg/L	Removal rate of sulfide reached 100%.	Chen et al., 2017
	CSTR (14 L)	primary sludge and waste activated sludge	0–6 air volume/g TS/min	Biogas production increased by 17.8%	Ruan et al., 2019
	EGSB (3.8)	Wastewater	Air aeration rate of 20 mL/min	Removal rates sulfide and nitrate reached 100% and 53.8%.	Zhang et al., 2019a
Pilot scale	CSTR+SOU (92 L + 1 L)	High-solid wastewater	7.2 L/d	Remove >99% of H_2S.	Duangmanee et al., 2007
	CSTR (250 L)	Sludge	oxygen flow of 0.013–0.024 L/Lreactor d	Remove >99% of H_2S.	Fdz-Polanco et al., 2009
	Pilot-scale plant (250 L)	Sludge	Oxygen flow rate of 0.2–0.25 N m^3 O_2/m^3 feed	Remove >99% of H_2S, with a final concentration of ~55 mg/N m^3.	Díaz et al., 2010
	Pilot reactor (256)	Sludge	O_2 dosage of 0.46 NL/L_{fed}	Biogas was successfully desulfurized (99%).	Ramos et al., 2012
	Anaerobic digester (50 L)	Primary sludge	ORP controlled (−320 mV to −270 mV)	H_2S in biogas decreased from over 6000 ppm to 30 ppm.	Nghiem et al., 2014
	CSTR (200*L*)	Sewage sludge	O_2 of 0.02%–0.20% (*v*/v)	All H_2S was rapidly removed from the biogas.	Ramos et al., 2014c
	Pilot-scale bioreactors (12.5 L; 25 L; 50 L)	Wastewater	DO level of 3.0–4.0 mg/L.	Sludge reduction efficiency increased from 42.9% to 68.3%	Niu et al., 2016
	Pilot-scale reactor (200 L)	Wastewater	Different amounts of air were dosed to determine the optimum air-to-biogas ratio	The content of H_2S decreased from 3000 ppm to <100 ppm.	Krayzelova et al., 2018
Full scale	Anaerobic digester (338 m^3)	Cow manure	Oxygen addition rate of approximately 1% of biogas production rate	Remove about 68.2% of H_2S.	Kobayashi et al., 2012
	Full-scale plants (1600–30,000 m^3)	Municipal wastewater	Air dosage of 0.28–6.0 m^3/h	The achieved desulfurization efficiency was 74% –99%.	Jenicek et al., 2017
	Full-scale plant (11,035 m^3)	Industrial wastewater and domestic sewage	DO of 0.5–1.0 mg/L	A low sludge yield of 0.074 g SS/g COD with high removal efficiencies of NH_4-N, TN, and COD.	Jiang et al., 2018

UAF - Up-flow anaerobic bio-filter; SOU - Sulfide oxidizing unit; UASB - Up-flow anaerobic sludge bed; LBR - Leach bed reactor; CSTR - Continuous stirred tank reactor; EGSB - Expanded granular sludge bed.

Jiang et al. (2018) reported the process performance of a full-scale SPRAS system by inserting a microaerobic tank and a clarifier before conventional activated sludge process. The wastewater treatment plant has a design treatment capacity of 20,000 m^3/d and maintains DO of 0.5–1.0 mg/L in the microaerobic tank. This full-scale SPRAS for industrial park wastewater treatment achieves a low sludge yield of 0.074 g SS/g COD with efficient removal of NH^4-N (98.2%), TN (77.2%), and COD ($SCOD_{eff}$ of 28.3 ± 9.3 mg/L).

The exploration on mechanism and effect in the microaeration-based AD processes are almost in a lab- and pilot-scale. At present, the full-scale application is lack of implementation. The conversion of outcomes obtained from pilot-scale studies into design of full-scale implementation needs further research and promotion.

6. Challenges and prospectives

Microaeration in AD system facilitates processing efficiency and system stability and has been applied in full-scale digesters. However, several challenges hinder the development and application of microaeration-based AD techniques. Further studies in the following aspects are needed to clarify the detailed mechanisms and pathway of organic degradation and promote the large-scale applications.

The automatic process control of microaeration is a challenge for implementation of industrial application. The exploitation of an automatic online data acquisition and supervisory control system for precise microaeration dosing is necessary for future studies.

Not all physiochemical parameters can be tested online. Thus, the research of composition and behavior of microorganisms is beneficial to improve performance. Evaluation of the microbial growth and substrate degradation kinetics in microaeration-based AD systems will be meaningful and instructive for further works.

Another important factor hampering the extensive application of microaeration-based AD systems is the additional cost due to aeration device and energy consumption. Aeration treatment is nearly impossible in some restricted locations, such as an aeration treatment of domestic sewage in inland ships, because such treatment is subject to space and cost. In practical application, the oxygen reaches anaerobic digesters unintentionally by feeding and mixing. The possible effects of such microaeration are neither precisely quantified nor exerted in standard AD models. The degradation rule of organic matters and succession of microbial community under natural aeration conditions (without extra aeration) needs further studies.

7. Conclusions

Microaeration has been investigated as an effective approach to promote the performance of AD process by accelerating hydrolysis, maintaining low VFA concentration, scavenging hydrogen sulfide, facilitating diversity of functional microbes, and increasing biogas production. The efficiency performance is mainly associated with microaeration rate, aeration method, substrate type, inoculum types, and other environmental factors. The application of pilot- and full-scale level cases shows a stable treatment effect with microaeration in AD process. Future works on the automatic process control and theoretical and practical aspects of organic degradation and microbial community shift under natural aeration conditions are needed for further studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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